WI-001: UVM-100 UV LED Optical Module Assembly

Document No.	WI-001
Revision	A
Date	2026-11-05
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Assembly	UVM-100 UV LED Optical Module
Linked documents	PFMEA-001, CP-001, UVM-100 assembly drawing, UVM-001 Rev B, FIX-001, GAG-100

Purpose

Defines the assembly sequence and in-process inspection for the UVM-100 UV LED optical module. The acceptance criteria in this instruction are the ones referenced by PFMEA-001 and CP-001.

Safety and handling

- Wear a grounded wrist strap and work on an ESD mat. Verify both with the ESD tester at the start of each shift (CP-001).
- Handle the lens only by its edge, using finger cots or the vacuum pen. Never touch the optical surfaces.
- No adhesive is used in this assembly.

Parts

Item	Part No.	Description	Qty
1	UVM-001	Housing, UV LED module (Rev B)	1
2	UVM-002	LED board, UV 3535 on Ø20 MCPCB	1
3	UVM-003	Lens, plano-convex Ø22	1
4	UVM-004	Retainer ring (Rev B)	1

Tools and equipment

- FIX-001 assembly nest
- GAG-100 seat-depth gauge and master setting part
- ESD mat, wrist strap and ESD tester
- Air blow-off
- Magnifier
- Inspection light
- Finger cots and vacuum pick-up pen

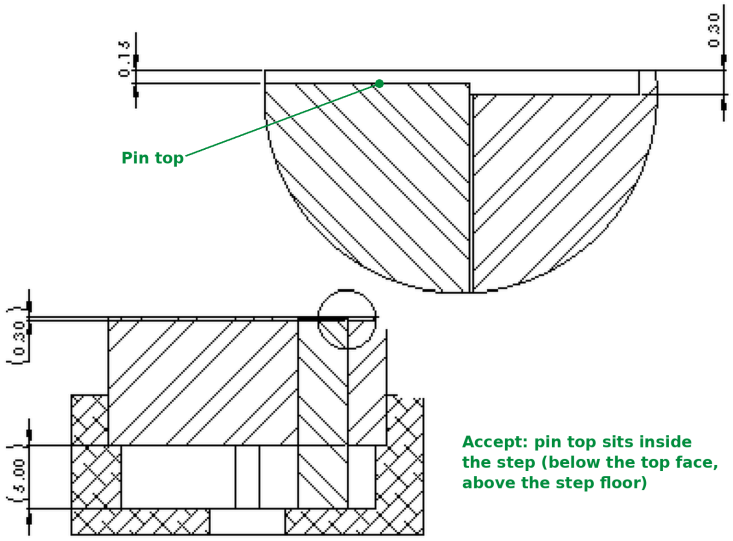
Before starting (each shift)

1. Check the ESD strap and mat with the ESD tester. Record the result in the ESD check log.
2. Check GAG-100 on the master setting part: the pin top must sit within the step. Record the result in the gauge verification log. If it fails, stop and notify Quality.

Assembly steps

OP10: Inspect housing seat depth

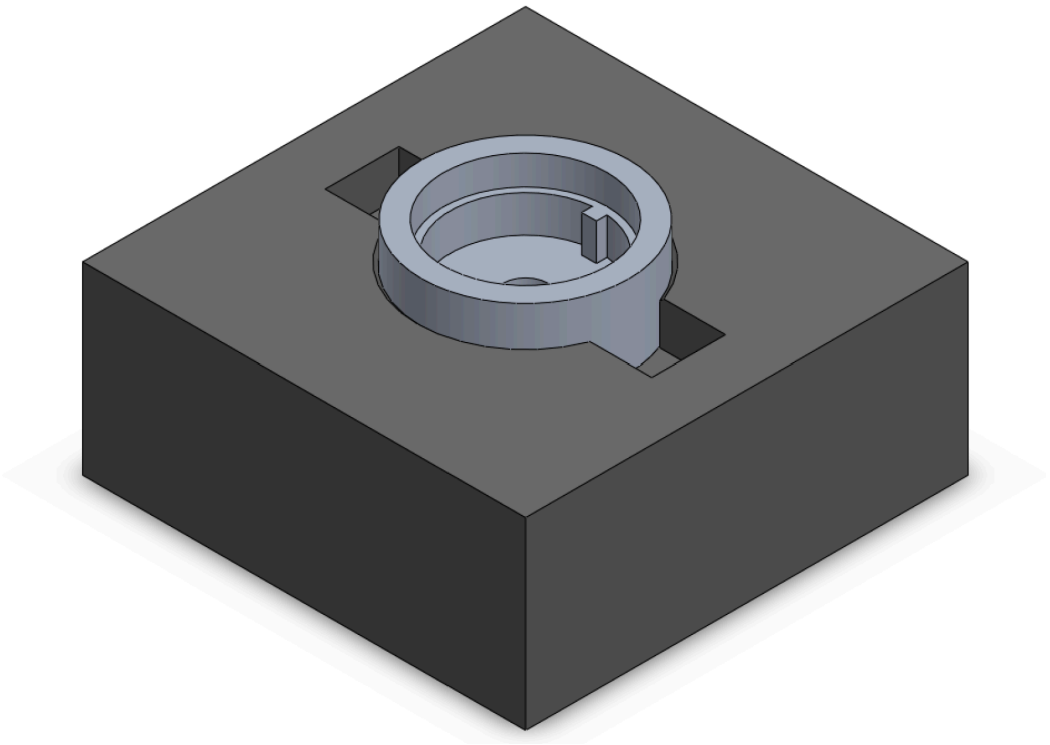
1. Place GAG-100 on the housing ledge, with the pin resting on the floor.
2. Check where the pin top sits relative to the step.



Accept: pin top is within the step (seat depth 5.00 ±0.10 mm).
Reject: pin top is below the step floor or above the top face. Quarantine the housing and notify Quality.

OP20: Load housing into fixture

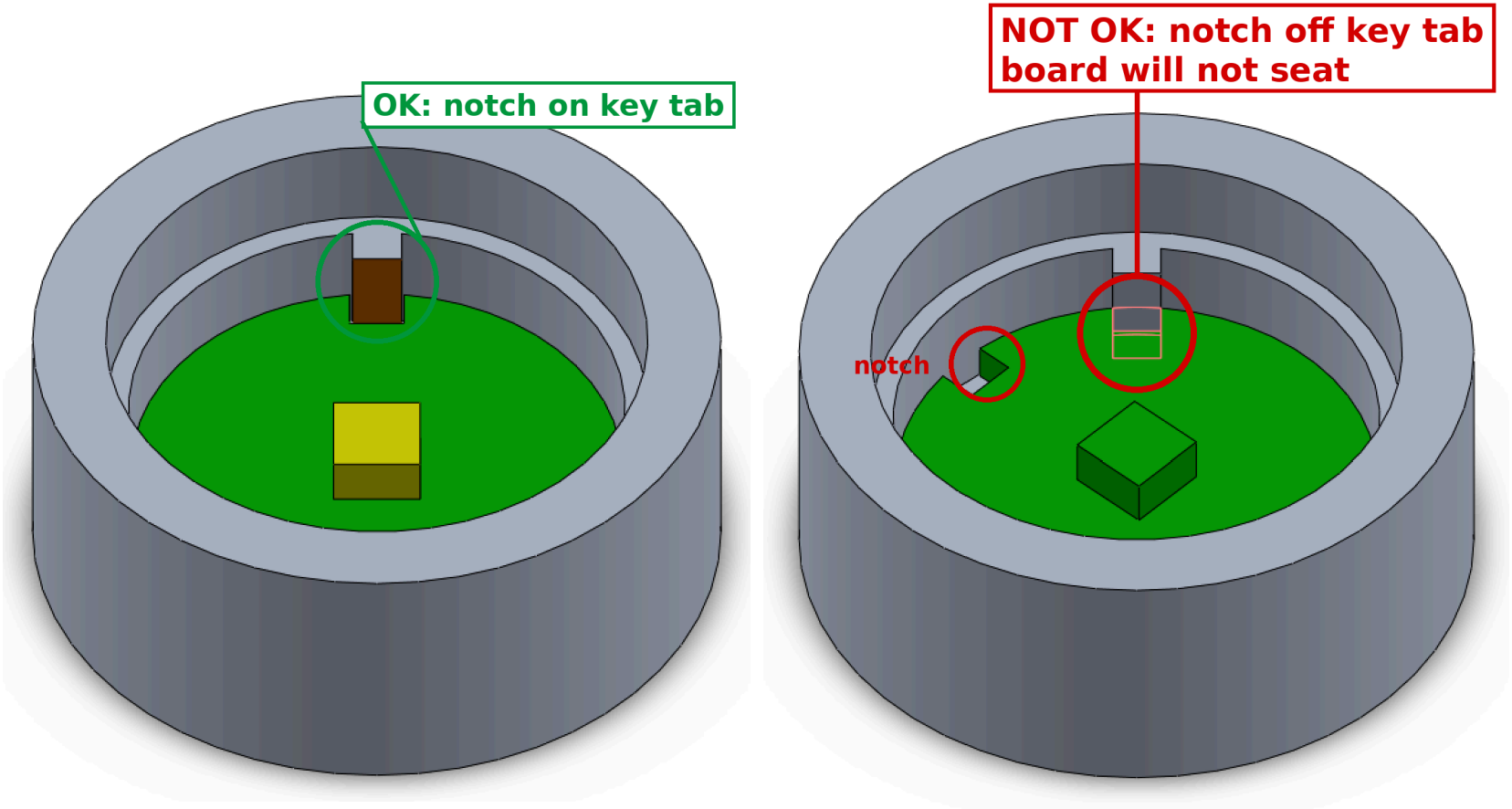
1. Check that the nest pocket is clean.
2. Place the housing in the nest pocket, open end up.



Accept: housing sits flat on the pocket floor and stands about 5 mm above the nest.
If not seated: remove it, clean the pocket and reload.

OP30: Install LED board

1. Blow off the housing floor with air.
2. Align the board notch with the key tab in the housing.
3. Lower the board flat onto the floor. Do not force it.



Key point: the board fits only one way. If it does not drop in, it is rotated. Do not press it.
Accept: board sits flat on the floor with no gap or tilt under the magnifier.
If not flat: remove it, clean the floor and reinstall.

OP40: Place lens

1. Blow off the ledge with air.
2. Pick up the lens with the vacuum pen.
3. Place it flat side down, dome up, onto the ledge.

Key point: the lens looks nearly the same both ways. Confirm the dome faces up before releasing it.

Accept:

- Lens rim sits flush and level on the ledge.
- No scratches, fingerprints or particles under the inspection light.

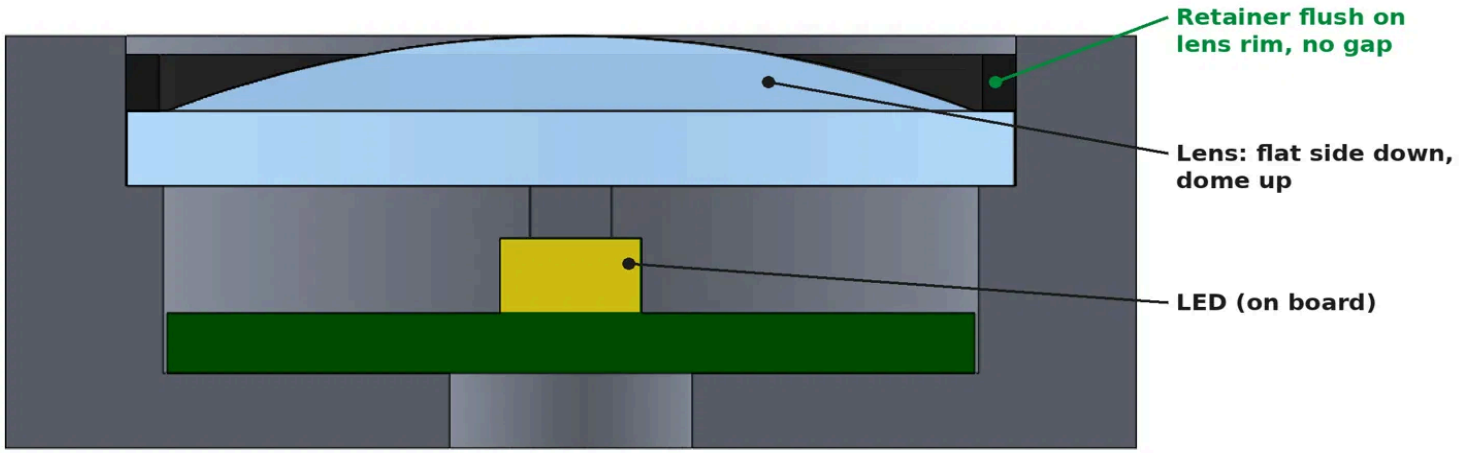
Reject: clean the lens if it is contaminated; scrap it if it is scratched.

OP50: Press retainer

1. Place the retainer ring on the lens rim.
2. Press it down evenly by hand until it seats. Do not use tools on the lens.

Accept:

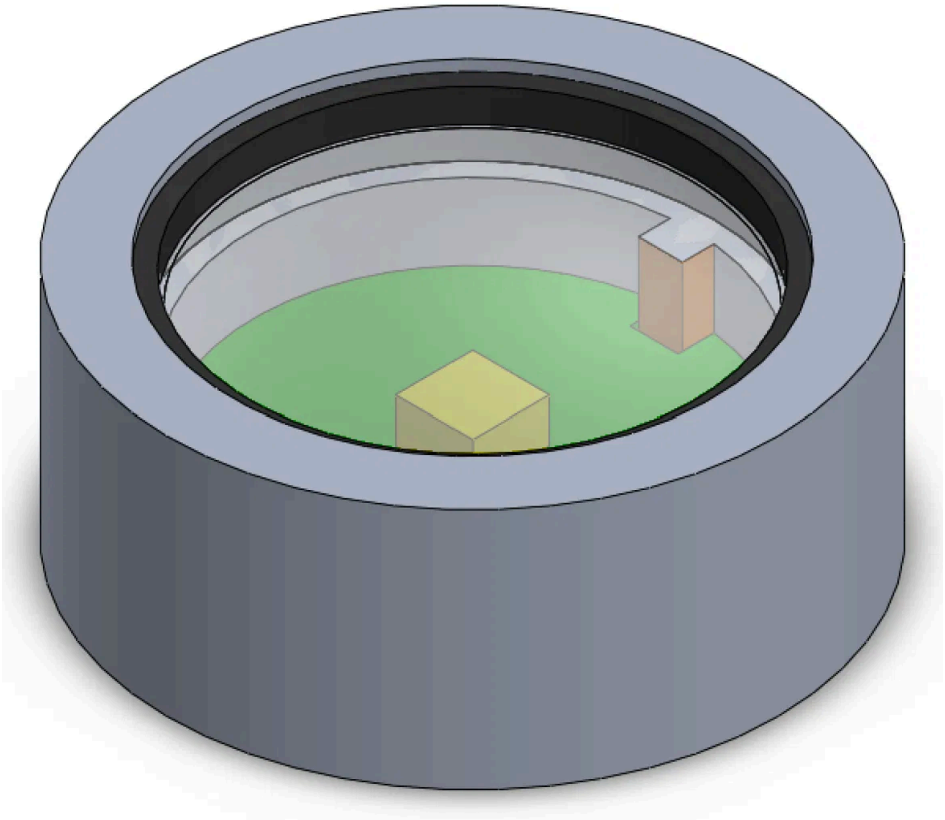
- Retainer sits flush and level on the lens rim, with no visible gap.
- No cracks or chips in the lens under the inspection light.



Reject: if the retainer is not seated, re-press it once; if it still does not seat, scrap the retainer. If the lens is cracked, scrap the module and notify Mfg Eng.

OP60: Final inspection

1. Inspect the module against the boundary samples.
2. *(Planned, per PFMEA-001 #5 and #12) Power on the module and confirm light output.*



Accept: matches the boundary samples and all OP10 to OP50 criteria are met.
Reject: quarantine the module and route it to rework or scrap.

Revision history

Rev	Description	Date	By
A	Initial release	2026-11-05	O. Dhinoja